

TEAM UPDATE: ISC2 x CYBLAC Cybersecurity Bootcamp Recap (May 25 - June 1, 2025)

Hello Team,

Our second week in the ISC2 x CYBLAC Cybersecurity Bootcamp (May 25 - June 1, 2025) was packed with critical insights to empower us in cybersecurity. Below is a detailed yet concise recap of the sessions, highlighting key events, essential cybersecurity concepts, and actionable next steps. With cyber threats like ransomware and insider risks rising, these lessons are vital for building resilient systems and advancing our careers.

Key Events and Session Highlights

Sunday, May 25: Incident Response, Business Continuity, and Disaster Recovery

Event: Sir Austin James introduced Incident Response, Business Continuity (BC), and Disaster Recovery (DR).

Key Topics:

- **Incident Response:** Defined as addressing violations of the CIA Triad (Confidentiality, Integrity, Availability) through five steps: preparation, detection, containment, eradication, and recovery. Tools like SIEM and forensic analysis aid detection, while containment prevents threat spread (blocking malicious IPs).
- **Business Continuity:** Ensures critical operations during disruptions via risk assessments and Business Impact Analysis (BIA). Backup sites (hot, warm, cold) and updated employee contacts are key.
- **Ransomware:** Never pay ransoms due to risks of no data recovery, legal issues, and further attacks.

Why It Matters: These concepts prepare us to mitigate threats like phishing and ransomware while ensuring organizations maintain operations during crises.

Wednesday, May 28: Incident Response, BC, and DR (Continued)

Event: Sir Austin continued with BC and DR, assigning a practical log analysis task and discussing recovery metrics.

Key Topics:

- **Recovery Metrics:** Recovery Time Objective (RTO) defines downtime limits, and Recovery Point Objective (RPO) sets acceptable data loss (hourly backups).
- **Backup Strategies:** Use full (weekly), differential (daily), and incremental (every 30 seconds) backups to minimize data loss. Off-site storage protects against regional disasters.

Case Studies: Target's 2013 breach showed poor network segmentation risks, while Maersk's recovery success highlighted the value of simulations and backups.

Why It Matters: Practical tasks and metrics like RTO/RPO ensure we can restore systems efficiently, as seen in high-stakes environments like Amazon's 9-second downtime SLA.

Saturday, May 31: Access Control Concepts

Event: Ma'am Maryam Shuaibu Aliyu introduced access control fundamentals, focusing on securing resources and preventing breaches.

Key Topics:

- **Access Control Principles:** Governs who accesses resources using least privilege (minimum access needed) and segregation of duties to reduce fraud.
- **Mechanisms:** Privileged access for admins, Data Loss Prevention (DLP), and immediate access revocation for ex-employees.

Case Study: MedShare's breach from cloned accounts highlighted the need for role-based access control (RBAC) and quarterly permission reviews.

Why It Matters: Proper access control prevents insider threats and data leaks, critical for protecting sensitive data like patient records.

Sunday, June 1: Access Control Concepts (Continued)

Event: Maryam explored physical and logical access controls, assigning a task to develop an access control policy.

Key Topics:

- **Physical Controls:** Include locks, CCTV, and biometrics to secure facilities. Tailgating prevention and vendor ID checks are key.
- **Logical Controls:** Use RBAC, multi-factor authentication (MFA), and access control lists to restrict system access.

Task: Teams to review task folder shared on ClickUp for violations and draft an ISO 27001-aligned policy emphasizing least privilege.

Why It Matters: Combining physical and logical controls creates a defense-in-depth strategy, safeguarding both data and infrastructure.

Key Cybersecurity Takeaways

- **Incident Response is Proactive:** Structured steps (preparation to recovery) and tools like SIEM ensure rapid threat neutralization. Ransomware payment risks (legal, reputational) underscore prevention over reaction.
- **BC and DR Ensure Resilience:** BC maintains critical functions, while DR restores systems using backups and recovery sites. Metrics like RTO/RPO and cases like Maersk show the value of preparation.

- **Access Control is Foundational:** Least privilege, RBAC, and segregation of duties prevent breaches, as seen in MedShare's failure. Physical (biometrics) and logical (MFA) controls work together for robust security.
- **Training Reduces Risks:** Phishing awareness and regular simulations (tabletop exercises) are critical, as human error remains a key vulnerability.
- **Career Relevance:** Skills in incident response, BC/DR, and access control are in high demand, aligning with the CC certification's focus on practical security.

Action Items and Reminders

- **Tasks (Due June 12):** No class from 2nd June to 13th June 2025.
 1. *Incident Response* Task available on ClickUp.
 2. *Access Control* Task is available on ClickUp.
- **Next Session:** Virtual Mentorship session on Saturday June 14, 2025 and Task presentation on Sunday June 15, 2025.
- **Engagement:** Attend live sessions (mandatory), mute mics when not speaking, use full names, and consult mentors for career guidance.

NOTE : Watch the recorded session for in-depth understanding if you missed the session.

Best regards,

Prince Kwesi Lassey (Team Lead) & Joachim Araga (Assistant Team Lead)